



Healthcare Mobile Application

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01

Overview

Final Description of Mobile Application



Healthcare Application



A centralised and secure mobile application for individuals to conveniently monitor, track, and log their health-related information.

- 24 CRUD APIs (14 Local Cached)
- Proper Navigation
- 2 Sensors: Accelerometer & Gyroscope
- Backend Test Scripts (19/24 APIs)
- GitHub Repo for both frontend & backend
- CI/CD via GitHub Actions
- Functions without Internet Access



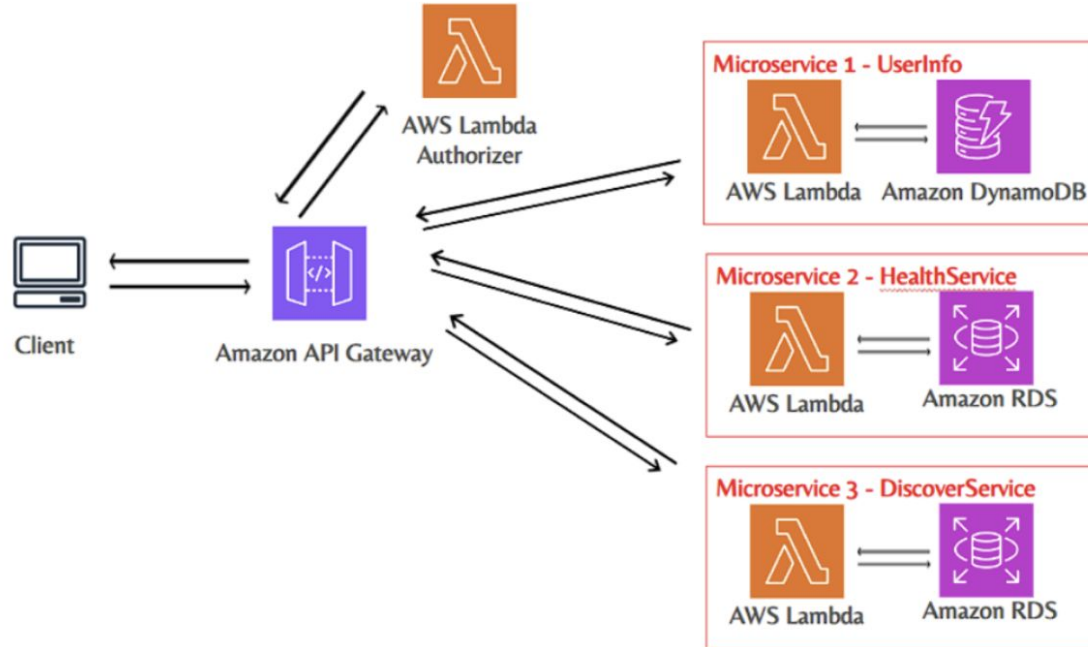


02

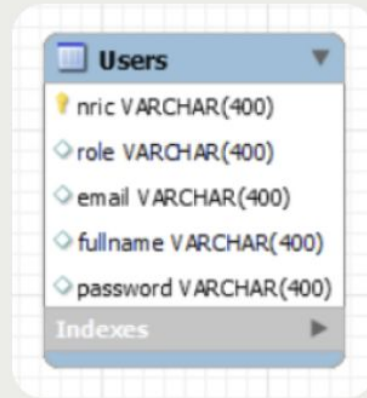
Backend

Explanation of Backend System

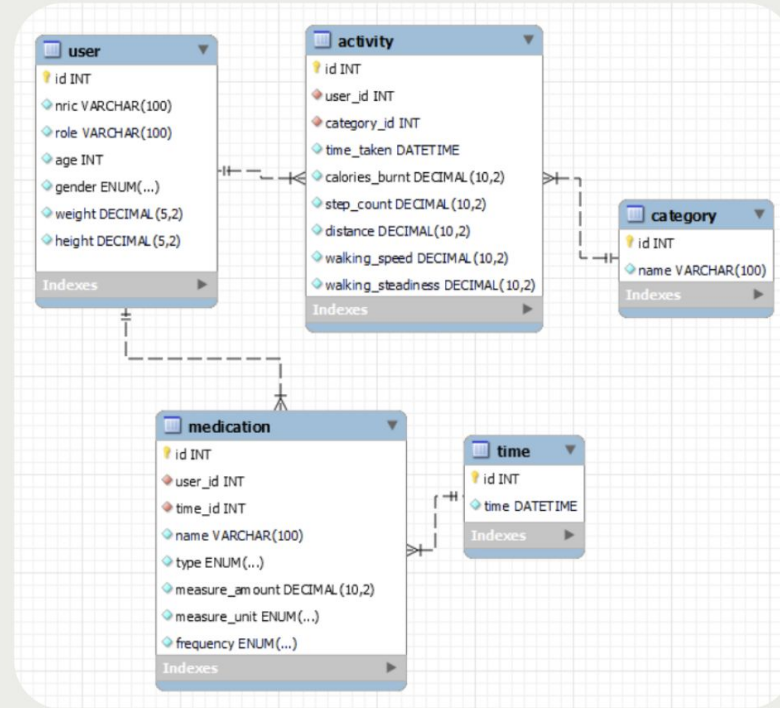
Architectural Diagram



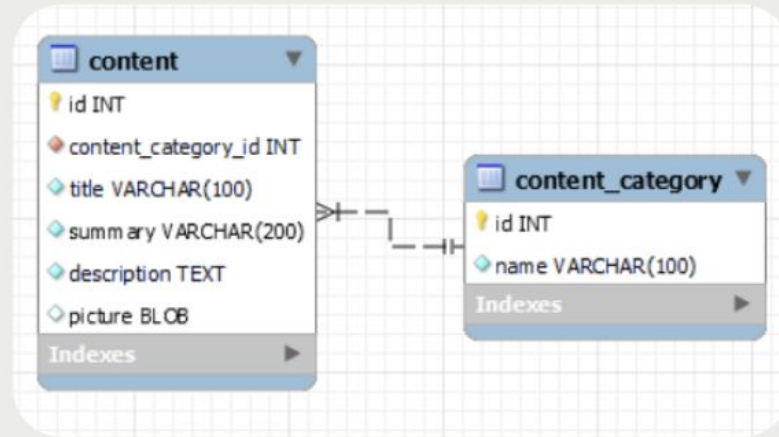
Microservice #1 – UserInfo



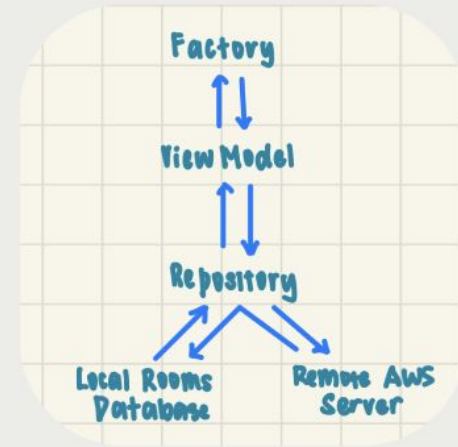
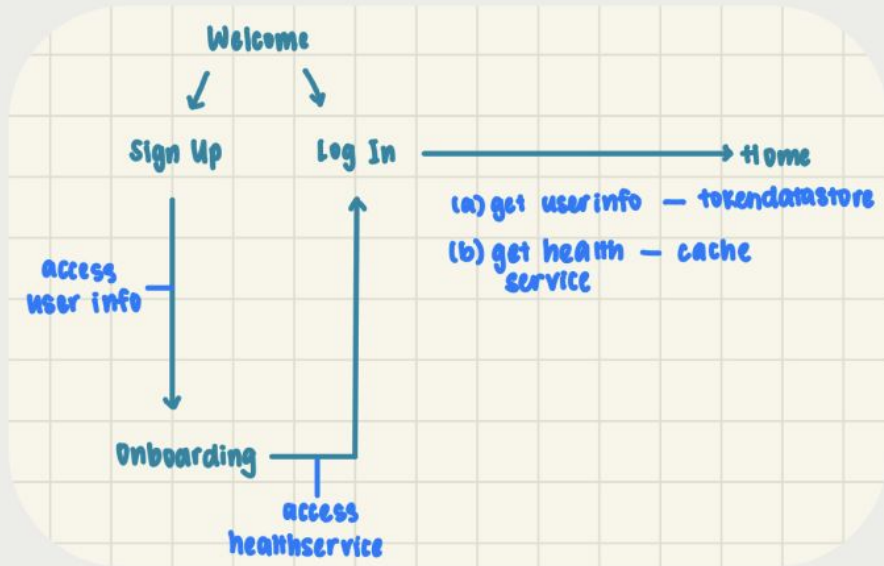
Microservice #2 – HealthService



Microservice #2 – DiscoverService



Backend Flow System





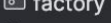
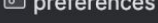
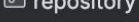
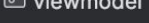
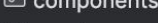
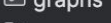
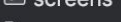
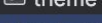
03

DevOps

Explanation of DevOps System

Repository Structure

- ▼  .github/workflows
 -  node.js.yml
- >  .idea
- ▼  backend
 - >  lambda_authorizer
 - >  lambda_discover_service
 - >  lambda_health_service
 - >  lambda_user_info
- >  frontend
 -  .DS_Store
 -  README.md

- ▼  app
 - ▼  manifests
 -  AndroidManifest.xml
 - ▼  kotlin+java
 - ▼  com.example.fullhealthcareapplication
 - ▼  data
 - >  dao
 - >  database
 - >  entity
 - >  factory
 - >  preferences
 - >  repository
 - >  viewmodel
 - ▼  ui
 - >  components
 - >  graphs
 - >  screens
 - >  theme
 -  MainActivity.kt
- ▼  com.example.fullhealthcareapplication (androidTest)

CI/CD System

```
PASS test/lambda.test.js
GET /info/{nric}
  ✓ Should return a User based on NRIC (1 ms)
POST /user
  ✓ Should insert a new User and return a Success Response
PUT /user/{id}
  ✓ Should update User measurements and return a Success Response
DELETE /user/{id}
  ✓ Should delete a User and return a Success Response
GET /activity/user_id/{day}
  ✓ Should return a list of Activities for a User on a specific Day (1 ms)
POST /activity
  ✓ Should insert a new Activity and return a Success Response
PUT /activity/{id}
  ✓ Should update Activity details and return a Success Response
DELETE /activity/{id}
  ✓ Should delete an Activity and return a Success Response
GET /medication/user_id
  ✓ Should return all Medications for a User
POST /medication
  ✓ Should insert a new Medication and return a Success Response (1 ms)
PUT /medication/{id}
  ✓ Should update specified Medication details and return a Success Response
DELETE /medication/{id}
  ✓ Should delete a Medication and return a Success Response
GET /category
  ✓ Should return all Categories (1 ms)
GET /time
  ✓ Should return all Time Slots

Test Suites: 1 passed, 1 total
Tests: 14 passed, 14 total
Snapshots: 0 total
Time: 0.107 s, estimated 1 s
Ran all test suites.
```

```
PASS test/lambda.test.js
GET /content
  ✓ Should return a list of Content (2 ms)
GET /content-category
  ✓ Should return a list of Content Categories (1 ms)
POST /content
  ✓ Should insert new Content and return a Success Response (1 ms)
PUT /content/{id}
  ✓ Should update the specified Content and return a Success Response (1 ms)
DELETE /content/{id}
  ✓ Should delete the specified Content and return a Success Response

Test Suites: 1 passed, 1 total
Tests: 5 passed, 5 total
Snapshots: 0 total
Time: 0.278 s, estimated 1 s
```

```
- name: Install dependencies for lambda_discover_service
run: cd backend && cd lambda_discover_service && npm install

- name: Run backend tests for lambda_discover_service using Jest
run: cd backend && cd lambda_discover_service && npm test

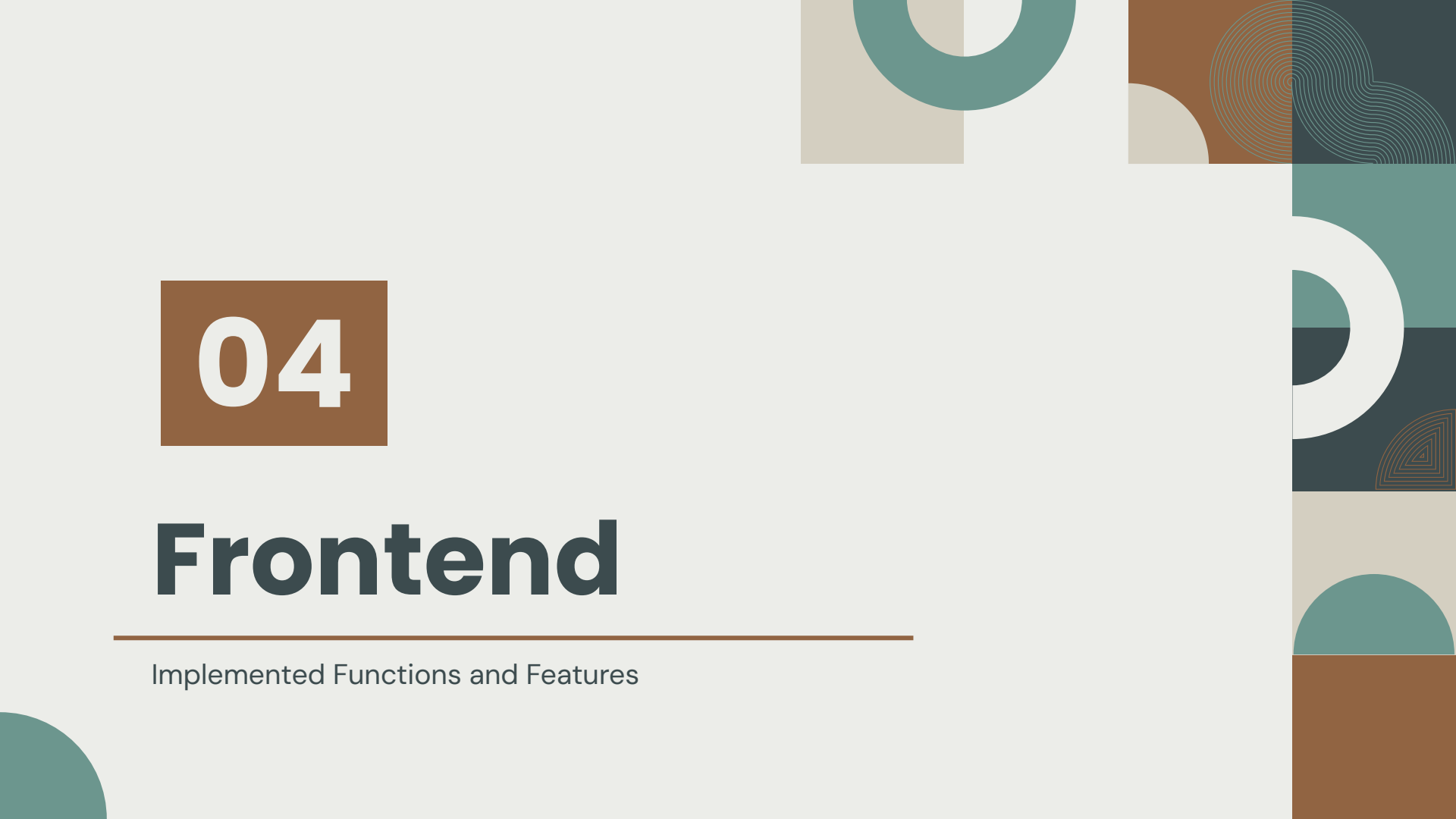
- name: Install dependencies for lambda_health_service
run: cd backend && cd lambda_health_service && npm install

- name: Run backend tests for lambda_health_service using Jest
run: cd backend && cd lambda_health_service && npm test
```

✓ Update Backend Testing Configurations	Automatic Testing for Backend #9: Commit a171071 pushed by sofiaamihan	main
✓ Implement Backend Test Scripts for lambda_health_service	Automatic Testing for Backend #8: Commit f3d4e70 pushed by sofiaamihan	main
✓ Update Discover Service index.js	Automatic Testing for Backend #7: Commit 96bba4d pushed by sofiaamihan	main
✓ Update Backend Testing Configurations	Automatic Testing for Backend #6: Commit ee47b62 pushed by sofiaamihan	main
✗ Update Backend Testing Configurations	Automatic Testing for Backend #5: Commit 2a81e6d pushed by sofiaamihan	main
✗ Update Backend Testing Configurations	Automatic Testing for Backend #4: Commit a029058 pushed by sofiaamihan	main
✗ Update Backend Testing Configurations	Automatic Testing for Backend #3: Commit b04ffa0 pushed by sofiaamihan	main
✗ Configure Backend Testing	Automatic Testing for Backend #2: Commit d055e3d pushed by sofiaamihan	main

Reflection



The slide features decorative geometric patterns in the top right and bottom left corners. The top right corner includes a teal circle, a brown square, and a dark blue square with concentric circles. The bottom left corner shows a teal circle. The bottom right corner has a dark blue square with concentric circles, a brown square, and a teal circle.

04

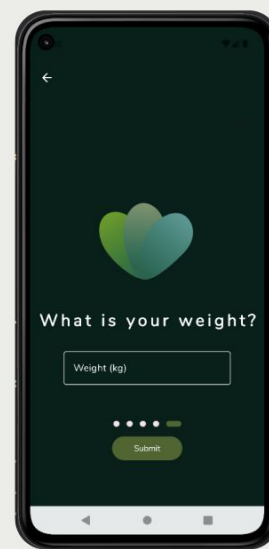
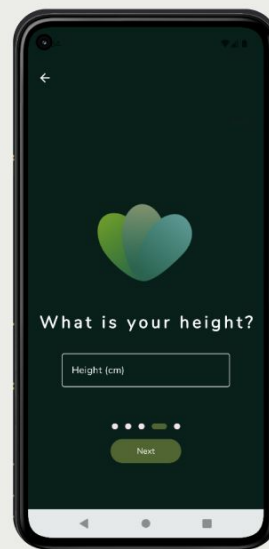
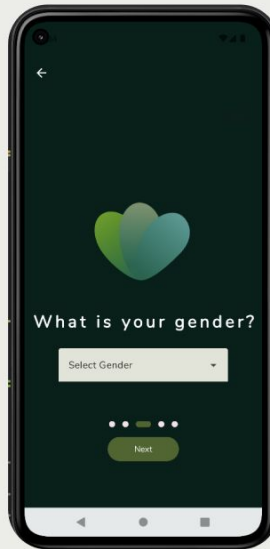
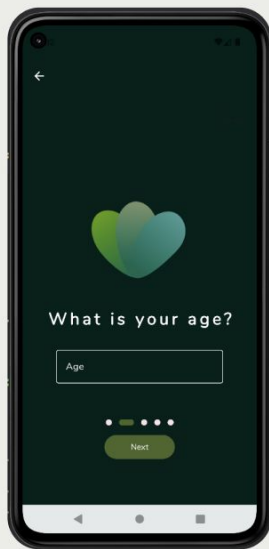
Frontend

Implemented Functions and Features

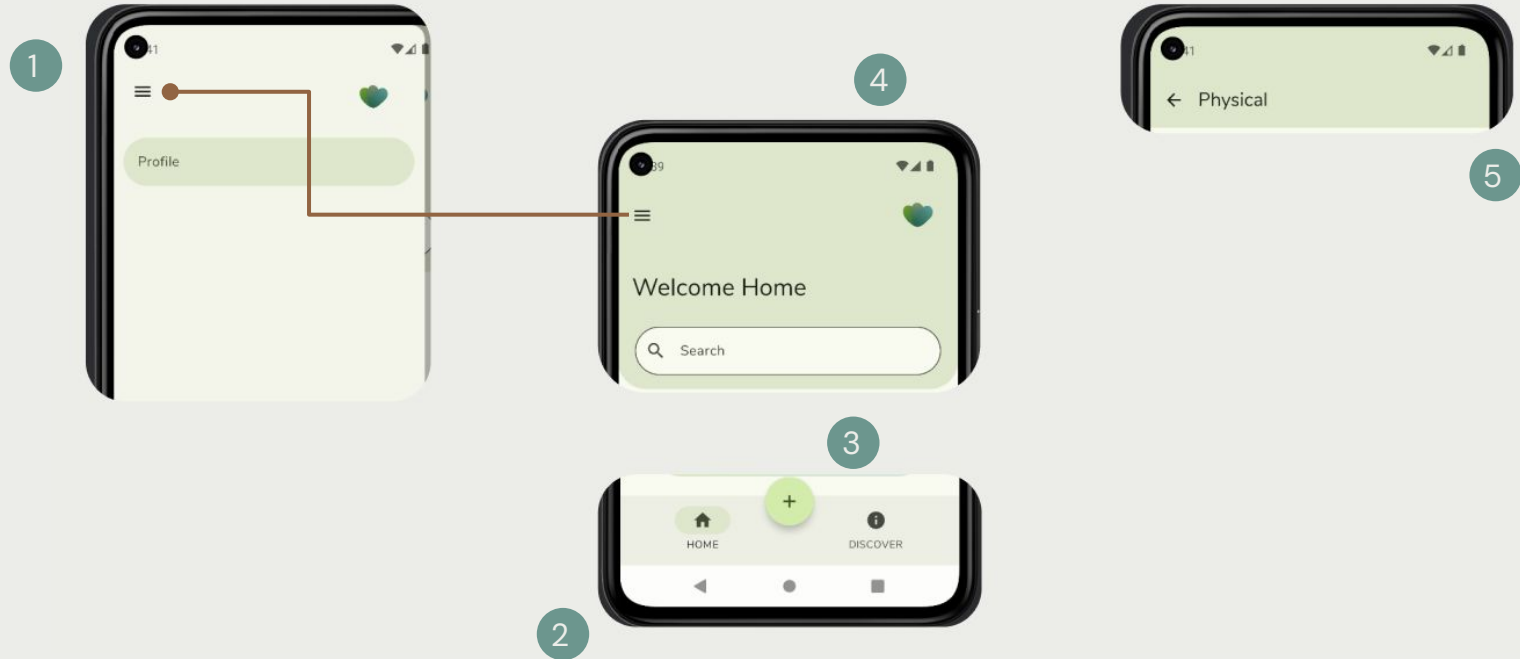
Authentication System



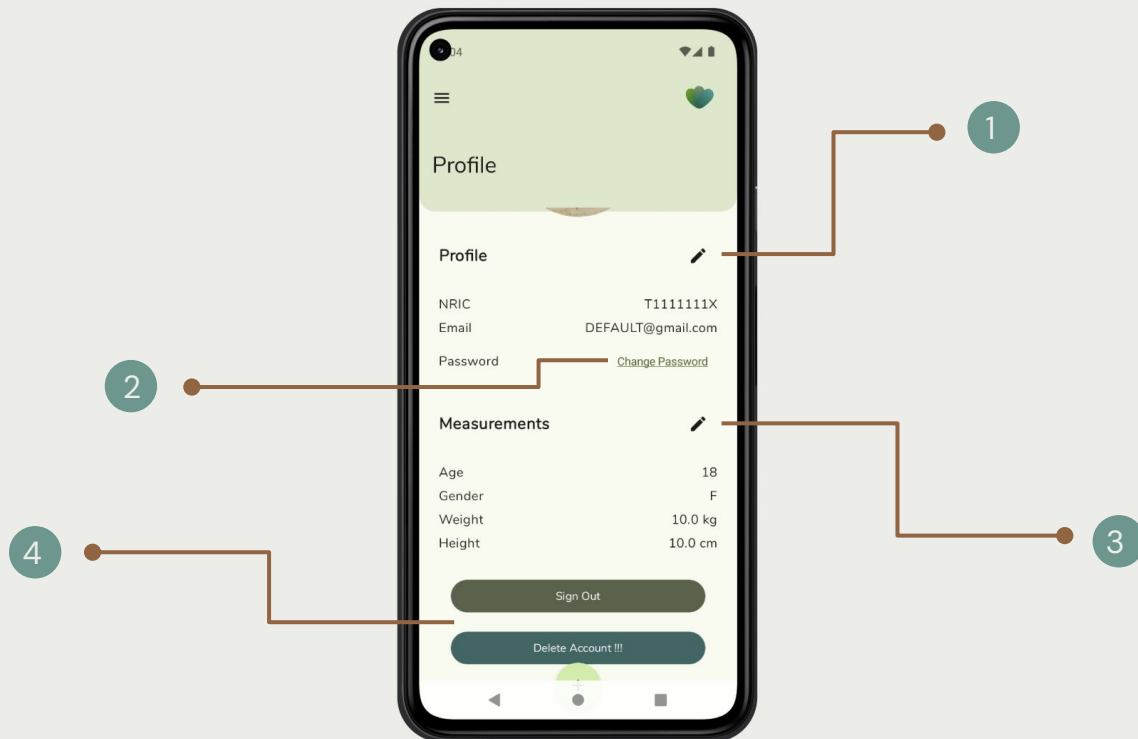
Onboarding Process



Navigation System



Profile Customisation



Sensors and Graphs



Utilities

 Search

All ▼

All

Diet

Exercise

Medicine

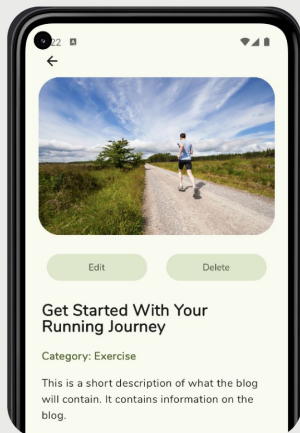
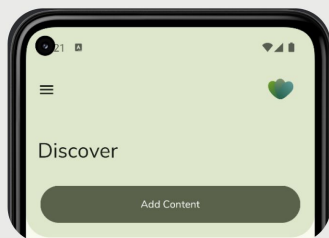
Mental Health

Environment

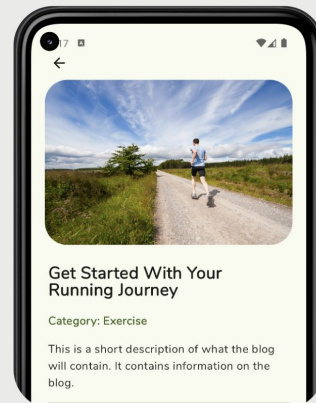
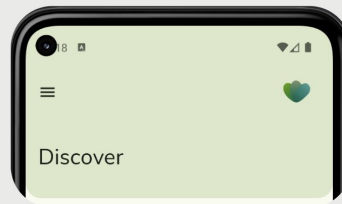
Disease

Locked Elements

Admins:



Users:



Features



Components





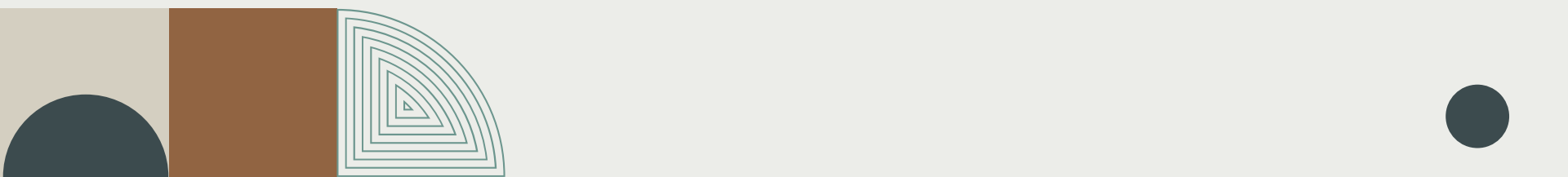
05

Reflections

Final Project Reflections



Learning Points –
What went well –
What did not go well –
Modifications from Original –
Improvements Moving Forward –





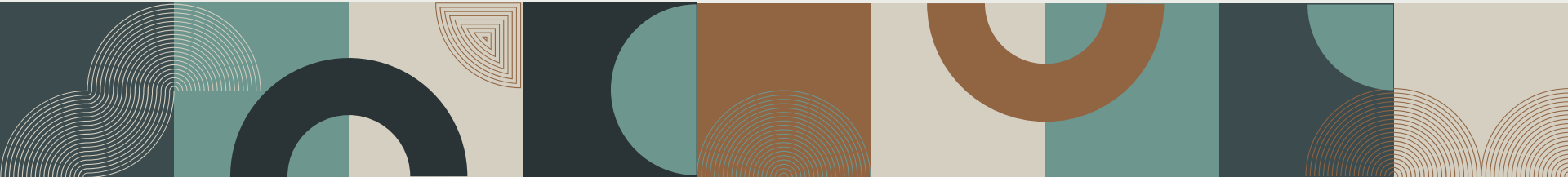
06

Demonstration

Demonstrating the Mobile Application

The top-left corner features a series of concentric, wavy lines in a light brown color, partially overlapping a teal square. The top-right corner contains a series of concentric squares in a light brown color, partially overlapping a brown square, a dark blue square, and a teal square. The text "Thank You" is centered in a bold, dark blue font.

Thank You

The bottom-left corner features a series of concentric, wavy lines in a light brown color, partially overlapping a teal square. The bottom-right corner contains a series of concentric squares in a light brown color, partially overlapping a brown square, a dark blue square, and a teal square.